

CALRIG · DESIGN DECISIONS REGISTER

CalRig design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items gathered from the review note, the decision records and the build work

CalRig design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review the design-for-construction changes P1 to P12 (front frame and four latches, door panel fixing, heat pump clamping, fan spacers, tray legs, mast, air-loop bulkheads and pumps, HEPA bulkheads, back-wall glands, drip tray and drain, aerosol bulkhead, jar rack) and the 9 mm base, made under Amish's 2026-09-30 instruction to make the design physically buildable	Accept; or change any item	Accept	The whole build plan	CLR-DDR-003, Tables 1 and 2
2	Mass margin: 13.8 kg against the 14 kg of R13, a 0.17 kg margin on estimated masses	(a) accept and weigh the prototype at TRL 4; (b) save more mass now, for example a 5 mm acrylic top	(a)	Section 5 mass check	CLR-DDR-003, A2
3	Base plate material: the appearance model shows dark HDPE, which at 12 mm would take the rig to about 15.8 kg	(a) sealed 9 mm birch plywood, as modelled; (b) HDPE with R13 relaxed again; (c) 6 mm HDPE on more feet	(a); HDPE for renders only if preferred	Base plate (section 3.1)	CLR-DDR-003, A3; REVIEW 2026-09-26, item 2
4	Collocation partner for the transfer SPS30 particle sensor (R8 not met until one is named)	A regulatory monitoring station, a university site or an AQ-SPEC style program	None yet	Not part of the build; needed before the SPS30 is used as the reference	CLR-DDR-001, O1
5	Appearance model items 1 and 3 to 9: faced, rounded jacket; front status light and name plate; printed door border; door panel pull handle; bubbler and dryer details; fan guards; example sensor heads; aerosol valve lever	Adopt for renders, adopt in the design, or drop, item by item	As recorded in the review note (mostly adopt for renders; decide on facing and status light at TRL 4)	Renders only, except the status light (controller wiring) and the pull handle (door panel)	REVIEW 2026-09-26, items 1 and 3 to 9
6	Report format that a city or funder would accept as calibration evidence	Open	None yet	Not part of the build; affects the fitting software (R14)	CLR-PRC-001, open questions
7	Larger sensor heads (HeatMap Node's 150 mm globe, SlopeWatch's capsule) that exceed the 90 x 70 x 50 mm bay size of R10	Add a "large item" case to R10 that takes two	None yet	Sensor tray bay layout (section 3.9)	REVIEW 2026-09-25, cross-repo notes

#	Decision needed	Options	Recommendation	Affects in the build	Source
		bays; or leave them to other rigs			

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The heat pump's sink bases are at least 120 x 110 mm and take four M4 clamp screws at 57 mm either side and 42 mm above and below centre	The bases clamp the wall over the 90 x 100 mm opening; the shell's clamp holes are cut to match	CLR-DDR-003, P3
2	An inner fin block and fan that reach about 0.20 K/W	R2 at 20 °C and 85 % RH depends on it	CLR-DDR-002, D7; CLR-PRC-001, open questions
3	The draw latches fit a 41 x 40 mm tab and reach a keeper 15 mm in from the tab edge	Sets the frame tab size and the keeper position	CLR-DDR-003, P1
4	The thread sizes of the bulkhead fittings (12 and 6 mm), cable glands (M20) and aerosol bulkhead (16 mm), and that the multi-hole gland inserts suit the sensor leads	The shell holes are laser cut to these sizes before anything is fitted	CLR-DDR-003, P7 to P11
5	The HEPA unit's ports are 12 mm and 80 mm apart at 117 mm above the bench, or can be drilled to that	The back-wall fittings run straight into them	CLR-DDR-003, P8
6	The SHT45 tolerance the supplier states (0.1 or 0.2 °C)	R5 is at risk if it is 0.2 °C	CLR-CAL-001, [F1]
7	The bimetal switches and relay: opening temperatures of 50 and 70 °C and a relay rated 10 A at 12 V direct current	R16 rests on them	CLR-DDR-003, Table 2

■ Value engineering

Value-engineering target: USD 412 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 439 (USD 27 over the target).

Main cost drivers (CLR-CAL-001, K): the reference cluster (line 8, \$132), the chamber shell (line 1, \$51), the Peltier heat pump (line 5, \$40) and the parts added for construction (\$27, of which new line 19, bulkhead fittings, cable glands and drain, is \$14). The CO2 path (SCD30 in the reference cluster and soda lime, line 16, \$8) is the largest block that is not core to the temperature, humidity and particle functions.

Savings worth trying:

- A core version without CO2 (no SCD30, no soda lime), about USD 372; this changes what the rig does and would be proposed, not made (CLR-DDR-003, A1).
- Re-pricing the fittings, glands and drain (line 19) and the front frame, drip tray and spacers (\$6) at purchase.
- Looking for savings in the reference cluster and shell; no single line obviously offers \$27.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items A1 to A6: budget raised to \$400, transfer references checked at salt and ice fixed points, accept the cooling limit (R1 redefined), incense smoke decay, carbon dioxide optional and nitrogen dioxide by field collocation only, insulated door panel	Amish: "i accept all your recommendations, go with them across all repos."	CLR- DDR-001, CLR- DDR-002
2026-09-25	Larger inner fin block (about 0.20 K/W); R13 relaxed to 14 kg; certified temperature probe only if the budget allows	Amish, same instruction	CLR- DDR-002, D7 to D9
2026-09-26	Budget set to \$412 to cover the priced BOM as it then stood	Amish: "i approve all the budget items."	CLR- DDR-002, O2